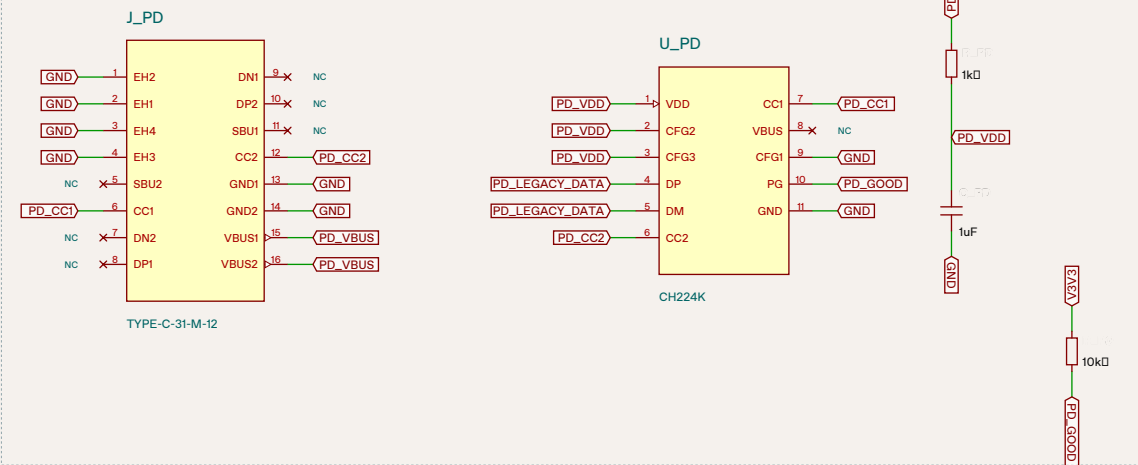


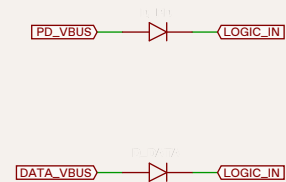
NEMA14 RP2040 | USB-C PD & Logic Power

15 V USB-C PD negotiation



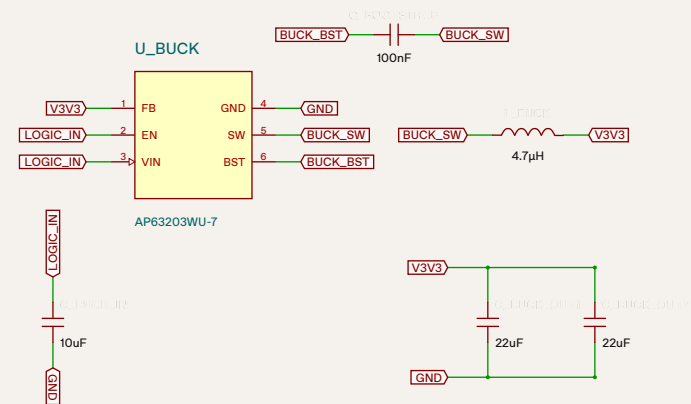
U\_PD / CH224K requests 15 V (CFG1=0, CFG2=CFG3=1).  
Open-drain PG is active low. DP/DM short: PD-only mode.  
R\_PD + C\_PD supply/filter the negotiation IC.

USB / PD diode OR



Schottky diodes accept either supply and block crossfeed.

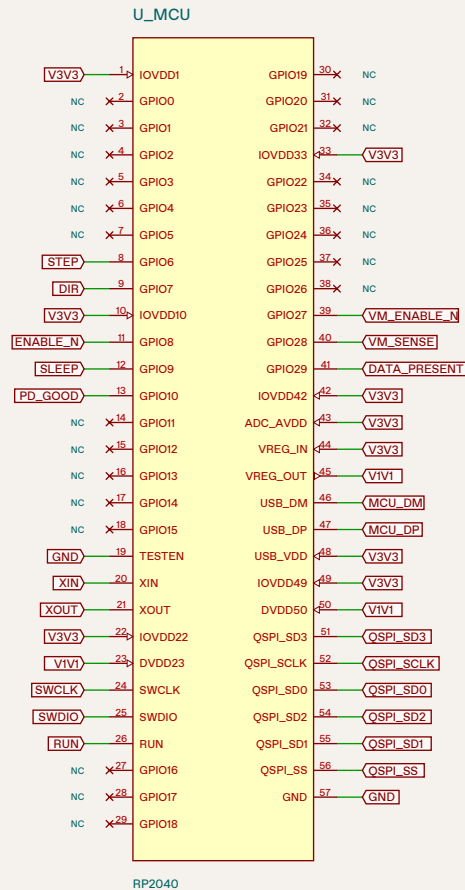
3.3 V buck supply



U\_BUCK / AP63203 converts LOGIC\_IN to 3.3 V.  
4.7 uH L\_BUCK and output capacitors filter the switching node.  
C\_BOOTSTRAP supports the high-side gate driver.

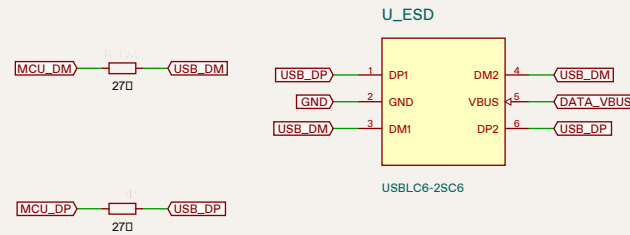
NEMA14 RP2040 | RP2040 & USB Control

Application MCU / physical pin map



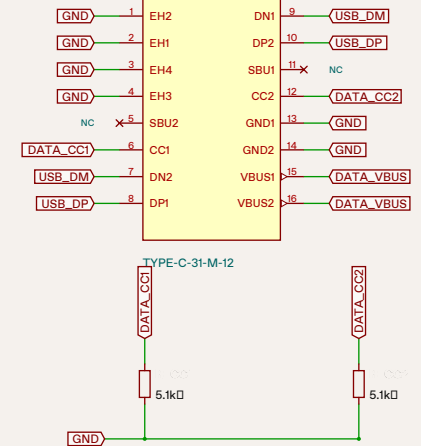
U\_MCU / RP2040 generates STEP/DIR and reads power status.  
Internal regulator: 1.1 V core. Flash/clock/bypass: next sheet.

Computer USB-C & ESD protection



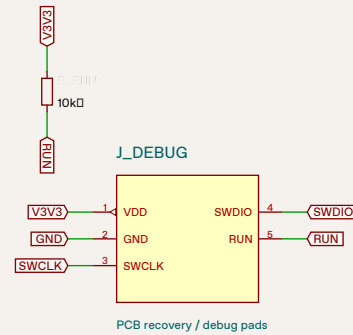
U\_ESD / USBLC6 protects D+/D-, R\_CC1/2 identify a USB sink.

J\_DATA

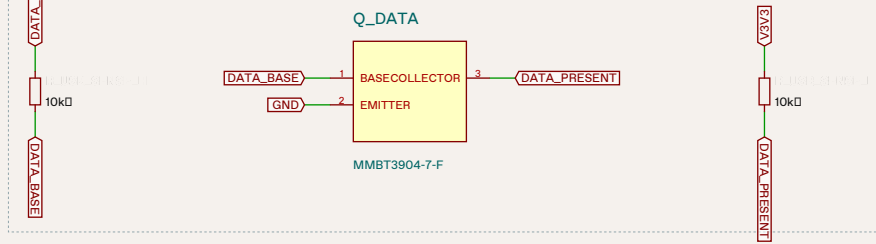


Programming & recovery

R\_RUN holds reset high; SWD / RUN pads allow debug.  
USB ROM boot: BOOT pads on next sheet.

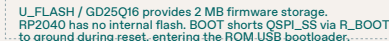


Host VBUS detection



Q\_DATA / MMBT3904 senses host 5 V without a direct GPIO feed.  
DATA\_PRESENT LOW means host attached.

## External QSPI firmware & ROM recovery



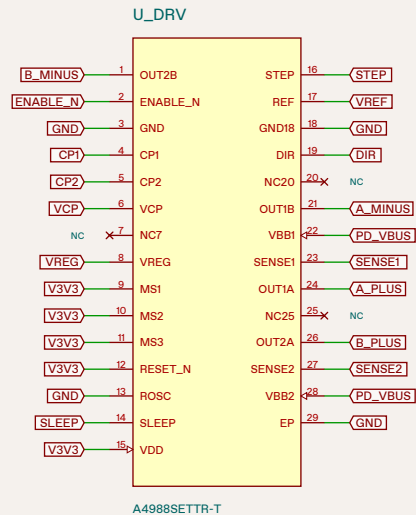
The schematic diagram illustrates the Y\_MCU module and its external connections. The module is a yellow rectangle labeled 'Y\_MCU' with four pins: 1 (XIN), 2 (GND), 3 (XTAL\_OUT), and 4 (GND). The module is identified as 'ABM8-272-T3'. The external circuit includes a 1kΩ resistor between XOUT and XTAL\_OUT, a 16.2MHz crystal between XTAL\_OUT and XTAL\_IN, and 15pF capacitors between XTAL\_IN and GND, and between XTAL\_OUT and GND.

Y\_MCU / ABM8 is the USB clock reference.  
Load capacitors support oscillation; R\_XOUT limits crystal drive.

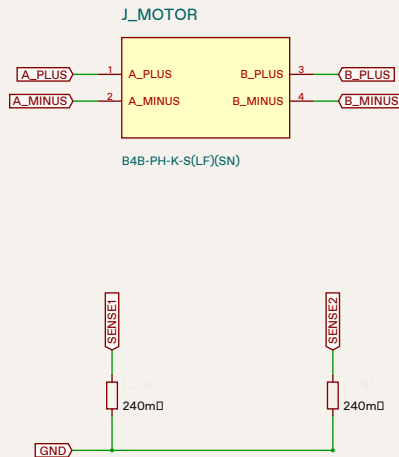
Each capacitor is a distinct fitted part.  
IO pin numbers appear in references.  
VREG\_OUT feeds VIV1; no external 1.1 V source.

NEMA14 RP2040 | Motor Driver & Voltage Monitor

A4988 / bipolar winding driver

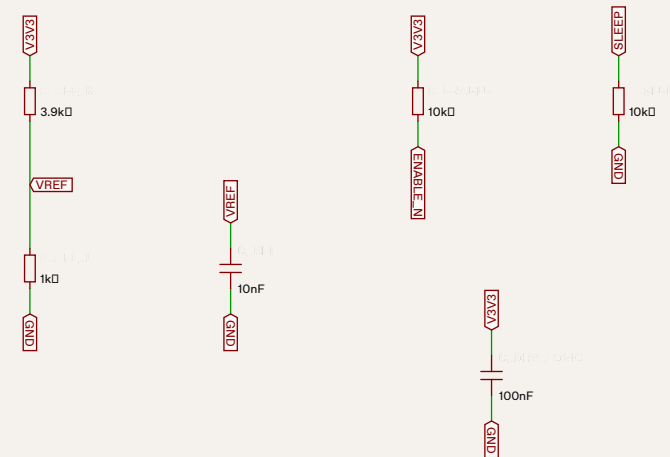


Keyed motor output & current shunts



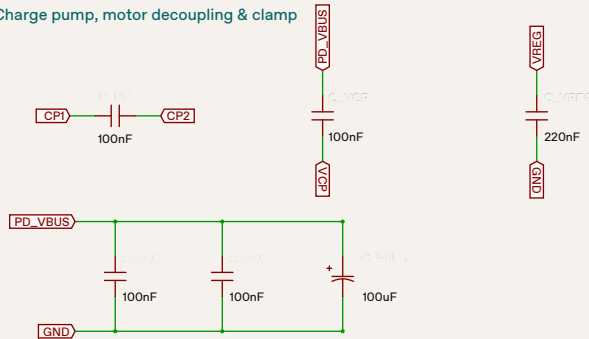
1 Black A+ 2 Green A-  
3 Red B+ 4 Blue B-

Current reference & safe defaults



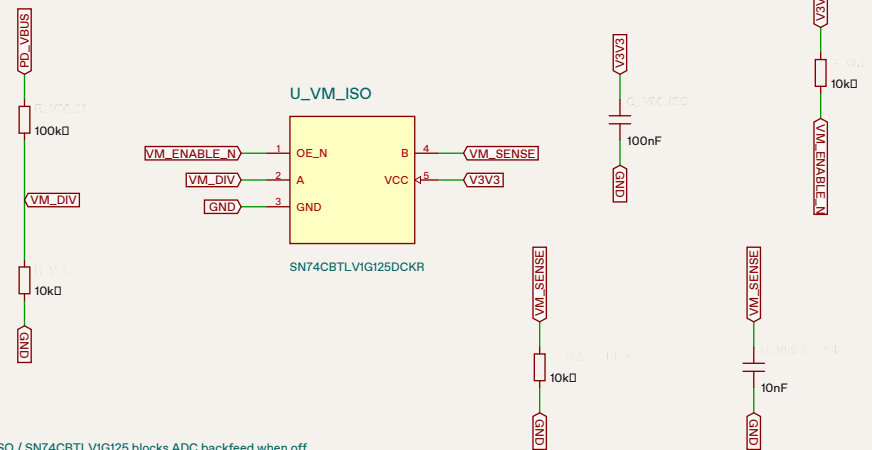
3.9k / 1k VREF and 0.24 ohm shunts: 0.351 A nominal.  
ENABLE pull-up / SLEEP pull-down keep the driver off.

Charge pump, motor decoupling & clamp



U\_DRV regulates both windings at fixed 1/16 microsteps.  
C\_CP / C\_VCP / C\_VREG support the internal driver rails.  
Bulk capacitance and TVS limit transients; regeneration needs bench tests.

Motor-voltage monitor / power-off isolation



U\_VM\_ISO / SN74CBTLV1G125 blocks ADC backfeed when off.  
OE pull-up defaults off; MCU enables, waits 0.5 ms, then samples VM/21.